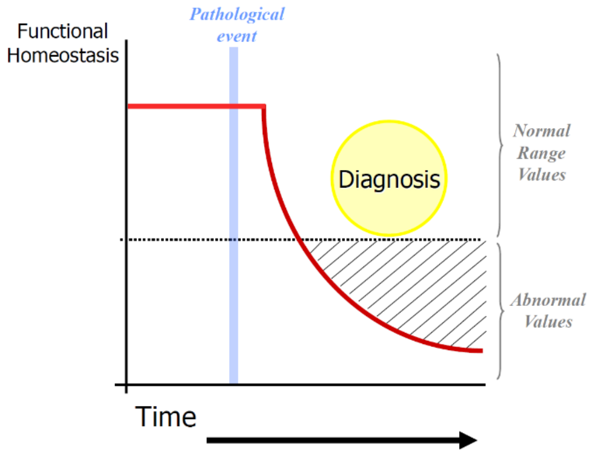
Emotion and behaviour

Humans maintain a healthy status in an active mode; we maintain a sort of equilibrium actively. At a certain point something happens in our body that will change reference values from a normal range to a pathological condition and on this phase, you make the diagnosis, based on something we can measure (symptoms).



What medicine has been found to do is to make diagnosis as close as possible to the moment in which something happens to the body. The moment in which you have a reasonable self-kindness that the patient has dementia or Alzheimer signs, the question is “How long before the dementing course has started?”, the answer is even 25 years.

The question we ask as engineers is “What can I do to detect a dementing process years before that the dementing process has led to a clinically evident proof”. This is called preclinical detection and is a huge challenge in medicine. This brings along important ethical questions, because detecting especially brain disorders does not imply that we have the tools and strategies to stop the onset of them, or even delay it, we have the impossibility to do anything from outside.

An example is the Huntington Chorea, neurodegenerative disorder characterized by the onset of uncontrolled movements with decay of cognitive functions. This disease becomes evident in 4,5,6 decades of life, but it is an autosomal dominant inherited disorder. Autosomal means that the gene is on the autosomal chromosomes, the ones that are not on the X and Y sexual chromosomes; When the disorder is link to an autosomal dominant gene, it means that all you need to manifest that disorder is just one gene. If you inherit the disease from one of your parents you have a 100% chance to develop the disorder, being it autosomal and dominant one is enough. If the gene is recessive, you got to have the two genes.

For disorders like this the ethical questions is whether having children being affected by it is ethic.

This introduction is to make clear why it is important to study the biological bases of brain function.

Searching for the mind

As for the case of Phineas Cage, the change in behaviour after the lesion in the frontal cortex confirmed that the frontal lobe must inevitably be important for controlling behaviour and decision making.

The ability to really enter into the brain and have a sort of true window to observe it in action and also to measure the brain structure, can be done by putting together different techniques. An important factor is molecular biology and genetics, since our genes shape our density.

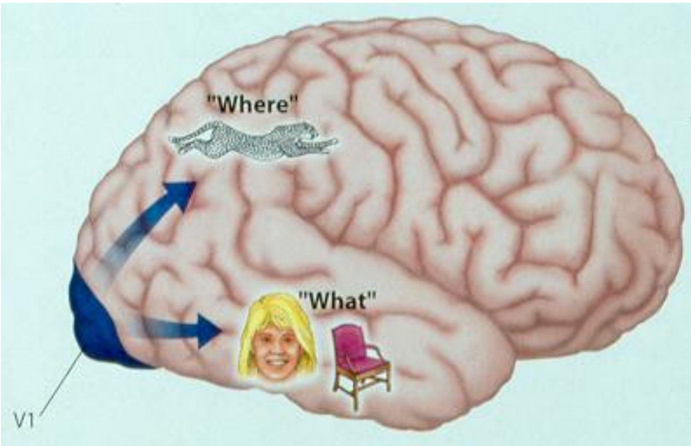
One aspect that in the future we will maybe deal with is drug resistant infections, which means that the ability of bacteria to change overtime is stronger than our ability to find new drugs. Each time they replicate errors happen by chance, the photocopy of their DNA may be with some mistakes developing resistance to drugs. This is the reason why we can't abuse with antibiotics.

Our brain is clearly different from the animals' ones, since our frontal cortex is much more developed.

With the development of MRI, PET scan, and all the techniques that allow us to see in the brain, I can also map what happen in the brain while doing some tasks. For example, in a face recognition task, you can measure accuracy and reaction time. With tasks like these, scientists have shown that once that the information has reached the back of the brain, the information gets divided in two parts, one to the ventral-temporal cortex (the "what" pathway) and the other one goes upwards to the superior part, the parietal-temporal area (the "where" pathway). In order for us to have a coherent representation of the external world we need to have an integration of information across different areas.

In the first stages of dementia, before you have a localized alteration within the brain, what happens is that the ability to dialogue between different areas gets lost.

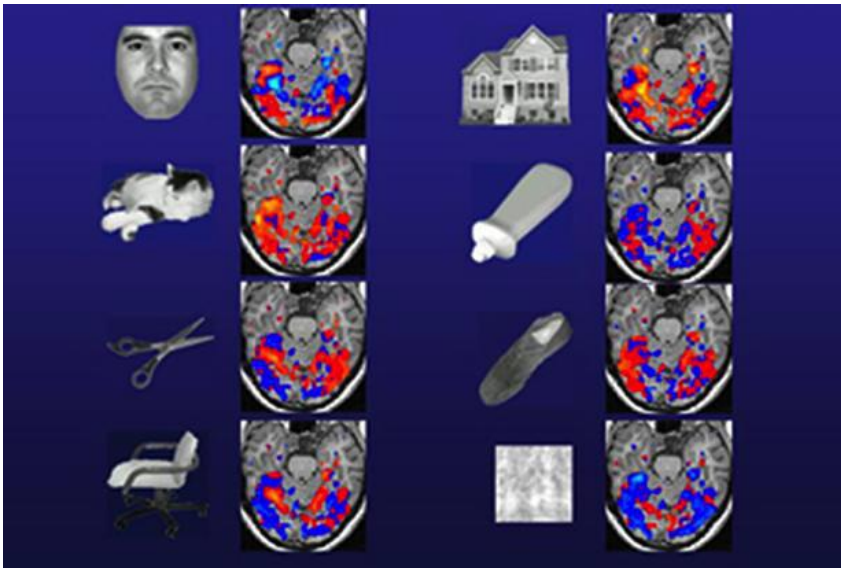
The ventro-temporal cortex is responsible of making us recognize an object from the external world.



But how can we distinguish an infinite number of objects relying on a finite part of the brain? There are different schools of thought, especially for body parts. Somebody says that, for example, for recognizing faces we have specialized groups of neurons. Somebody says also that there is an area for recognizing places, para-hippocampal place area. There can't be neurons for a specific object, there would be no space in the brain and evolutionary time.

MRI allowed to have a very good definition of the structure and see how the brain responds to different categories of object. It was found that you can't really say that for each specific category there is a certain and localized area responding, but there is a distributed engagement of the ventro-temporal cortex with a different pattern of response for the different objects.

If you compare the different brain responses, you don't find a localized response, but a distributed and overlapped response, with a different pattern of response. If you analyse how similar a response to the same category of object as compared to responses across categories you find that between categories that have a really strong correlation, brain responses are highly correlated. If the brain response is not specific, you get correlation within the same category as high as across categories.



There is a small statistical cavil, the correlation is not due to the distributed response but to some subset of area of the brain which drive it. If you have a subgroup with a very strong response, it can drive the correlation. So, what we did was excluding from the analysis the pixels with the highest response and see whether what was left would maintain the correlation or not, the response was yes. The brain does not respond all-or-none to a given category, but you have the ability to produce an infinite number of patterns of brain response, which are so specific for that given category that basically you can predict what an individual is looking at.

Is this pattern of brain response specific for vision or it's related to a more abstract representation and response. Is intended for the specific sensory modality or not? If you study subjects with projectile blindness, who have never seen but have a concept of things, and you ran a tactile study, what you find is that the response to the tactile task in bottle, shoes, mask recognition is a category specific response within the occipital retro temporal area (visual area). If you run in other subject and you do the visual and tactile study you get high correlation across modalities, which together do the sopra modal organization of the brain.

We can recognize the external world by producing in our given structures of our brain an input number of patterns. But when we look at the external world, we don't only recognize physical appearance but even before that we perceive and respond to the emotional and affective message that we have from a face.

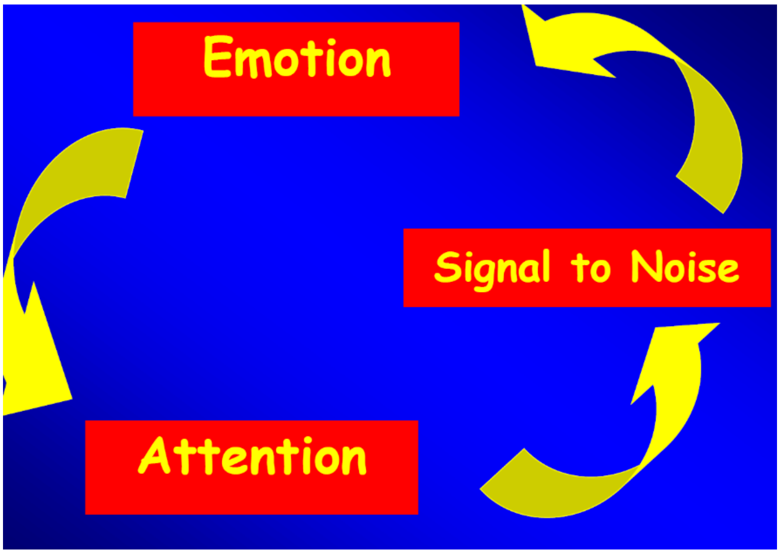
Emotions

The first thing that researchers did when these techniques of looking at the brain became available was to try to understand what happens when we see a terrified or normal face.

If you subtract brain activity for the neutral face to the one for the terrified one, you see the activation of the amygdales, two structures in the anterior medial part of the brain. They are the emotional computer of the brain, and if you modify these experiments even a little, by showing for example only neutral faces and then a scared face is shown for a very short period of time, the subject will say that he saw only neutral faces, but the amygdala shows a response in the exact moment the other face arrives.

The amygdala sends a signal downwards to the brain stem and the hypothalamus to increase blood pressure, the respiration, more oxygen to the muscles. The fear instinct.

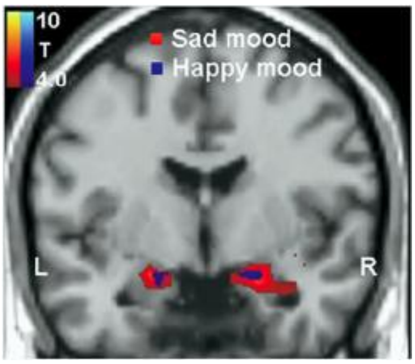
All you need to know is that there may be a danger to activate the chain of fear response, so that you are prepared in the case the danger is real. This health mechanism in the brain is very useful; the emotions increase the signal to noise so that you focus your attention on what you need to know. Emotions are instruments that evolution has equipped us with, in order to improve our survival chances.

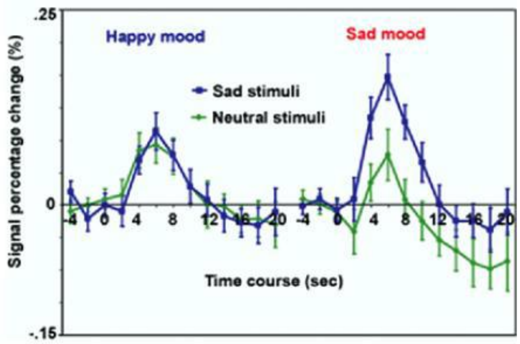


Our brain is programmed to filter what is not important. In the case of the photo of the twin tower you don't notice the individual changes in each photogram.

Emotions are also modulated by our affective state. It's important to know how mood modulates brain response to emotional stimuli.

Reaction time task → to people were shown different non-sense images, sad and neutral pictures, and brain activity was measured not only when they pressed the button at the arrival of the stimuli, but at each time. Researchers saw what happened in the brain when they saw a sad stimulus, compared to the neutral stimuli. But also, when subjects were already in the scanner bed, for the 15 minutes before starting the experiment it was projected a movie on the screen (funny or sad). What happened is that if you had been shown a happy movie, when you are shown a sad stimuli you see that you have a response to sadness in the amygdala, but you see that the





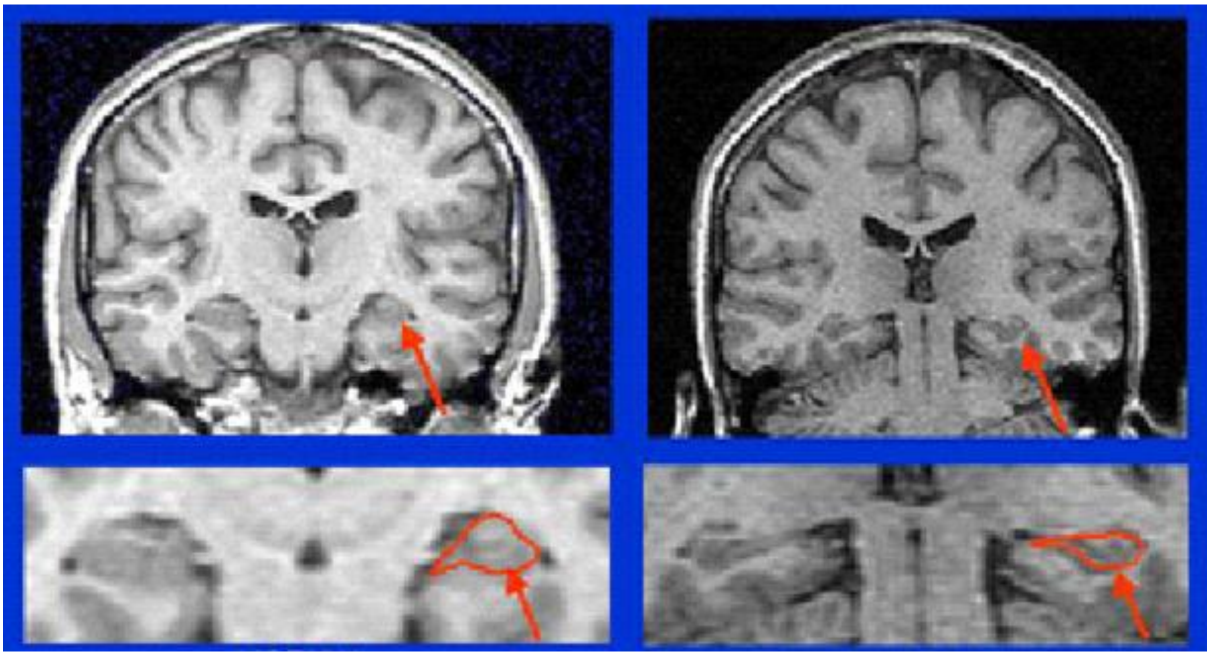
response is very limited in space and intensity. When you are shown the same picture, but you have watched the sad movie, you can see that the response is much larger within the amygdala and is much intense as compared to the other case.

15 minutes can determine a difference in brain response to the identical sad stimuli.

The take on message is that the way we perceive and react to external stimuli is not always the same, it can even change during the day.

The main risk factor for having a depressing episode is having the depressing episode itself.

We have seen over the last couple of decades is that depression is accompanied by alterations in the brain, specifically the atrophy of the hippocampus (is shrinked) in depressed people.



In brain disorders you have atrophy because you lose synapsis. Treatment of depression, for example, are associated with a disappearance of the shrinkage of the hippocampus, it's a reversible condition, and can be reversed not only by using drugs or therapies, but also by using meditation. There are better results when you do together psychic therapy and drugs.

If you practice meditation, you can have changes in brain structure.

Behaviour

How do we make a decision?

We can follow an instinct, an emotion, or we can evaluate and modulate that instinctual response by using reason.

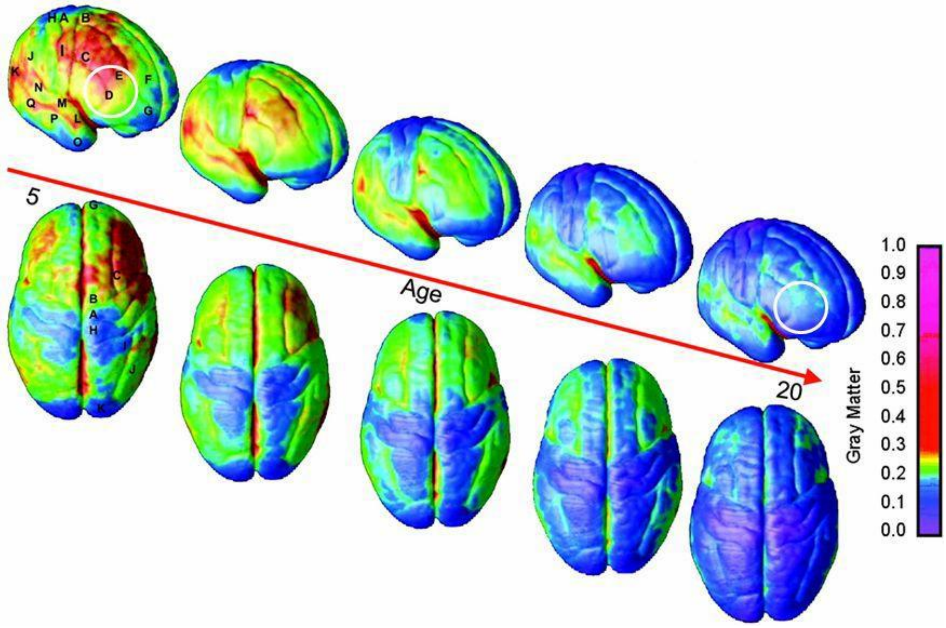
The bear doesn't eat the whole fish, he eats the skin only, because he is in search of the fat beneath the skin that he needs brown fat (it sustains metabolism for months) before hibernating. What he does is not a consequence of reasoning, he simply follows an instinct, an impulse that is genetically rooted into the bear habits.

But this doesn't mean that animals don't have reasoning, they have the ability to modulate the instinct.

How different can we be from bears?

We prefer calories rich food as compared to unfatty food because it’s written in our genetic code.

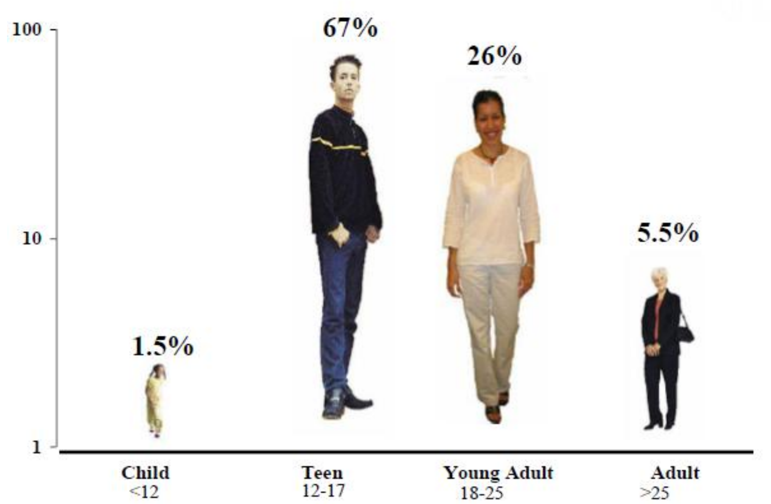
When we have genetically the right imprinted behaviour, we can control it better because of reason. Pre-frontal cortex has evolved in humans much more than in animals, and the pre-frontal cortex is the part of the brain, which is devoted to control behaviour, plan our action, decide what is good and bad, control our impulses. It is also true that pre-frontal cortex matures later in life.



We can see here, in colour scale, that the blue colour indicates the areas that have reached full maturation. At the age of five the primary visual cortex and the sensory and motor cortex are already very mature, the posterior part of the brain also matures very fast, while the anterior part of the brain doesn’t mature completely until the third decade of life.

So, our pre-frontal cortex is much slower in getting mature as compared to other parts of the brain, specifically to what we call the ancient part of the brain (limbic system), in which we process emotions, pleasure, sensation. Basically, there is a temporal shift between the

Dependence disorders begins in Adolescence



maturation of the ancient brain and the pre-frontal cortex, where we have behavioural control and decision-making. Adolescence is a period in which brain maturation happens; we still do stupid things.

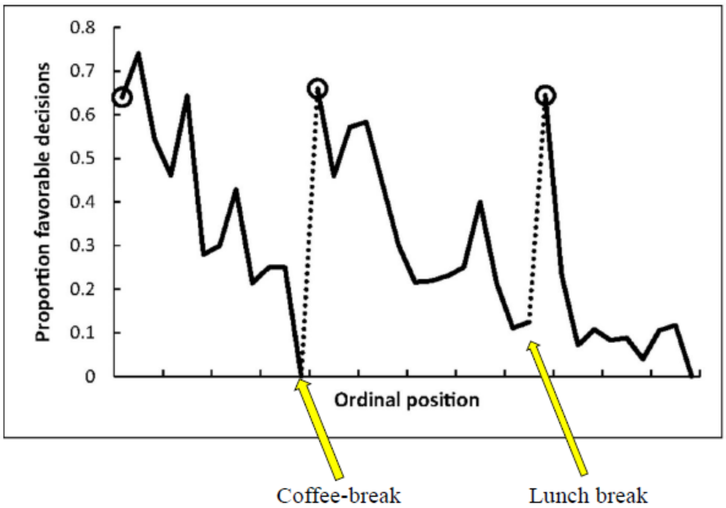
How much are we really in control of our behaviour?

A study tried to see if we can suppress spontaneous emotional responses. We are able to do it because there is a network in the brain which regulates emotional response that includes pre-frontal cortex.

In a study it was asked the subject to keep a neutral facial expression while they were shown a funny movie. What the researcher did was to provide all the subjects an actigraph the night before the experiment, that measures how long and how well you sleep. She tried to verify that the ability to suppress emotion (emotion suppression failures) is inversely related to sleep time. And she found out that the ability to control emotions is impaired when we are tired.

Does fatigue affect the decision-making process?

In a study, the researchers examined 1,112 judicial rulings by eight Jewish-Israeli judges and recorded what they have eaten during the day. For the results, the probability of getting a positive answer to the instance requested is very high in the morning, goes down processively, and reaches zero before the coffee-break. After that there is exactly the same cycle.



What happens in the brain when we are fatigued?

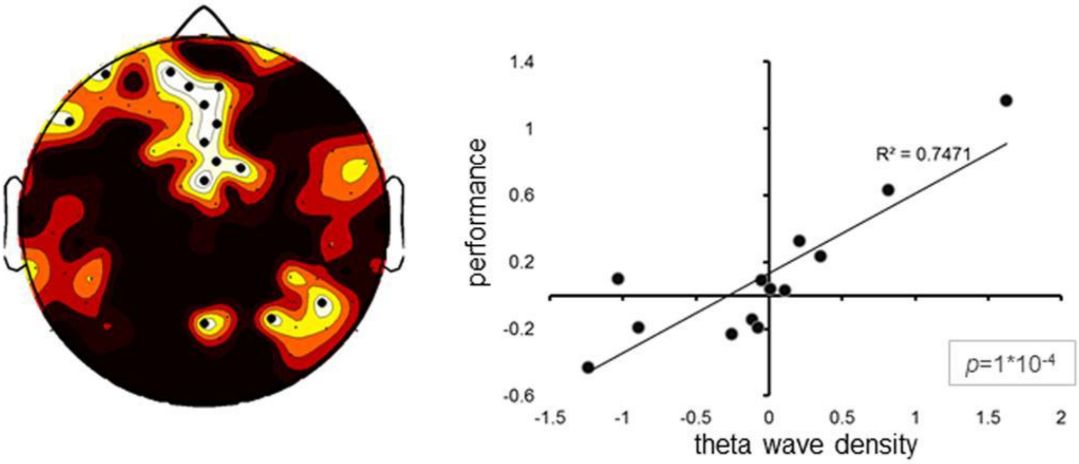
We naturally consider wakefulness and sleep as to mutually exclusive states, but this is not true. There is an intermediate condition between these two, that is called local sleep, which is a condition that happens while the individual is awake, but some regions of the brain go to sleep.

It was discovered that if you put a rat to get some food, at a certain point you can see that the rapid EEG rhythm changes at a certain point, becoming slower. They are phenotypically awake, but they can't reach the food. In coincidence with this missing there is a slow wave activity, not generalized all over the brain, but within the pre-frontal cortex, the area of the brain used the most.

Another study was devoted to see what happens when you do behavioural controlled tasks for a long time.

	Response	No Response
GO	Correct Response (CR)	Lapse (RT>500ms) (LA)
NoGO	Commission Error (CE)	Correct Withhold (CW)

What the researcher did was to get healthy young subjects and putting on them an EEG cap, making them perform the task for 24 hours. What was found is that if the subject presented doing the task significant episodes of slow wave sleep like EEG in the pre-frontal cortex, there was a linear correlation between the number of theta waves and the mistakes done in the task.

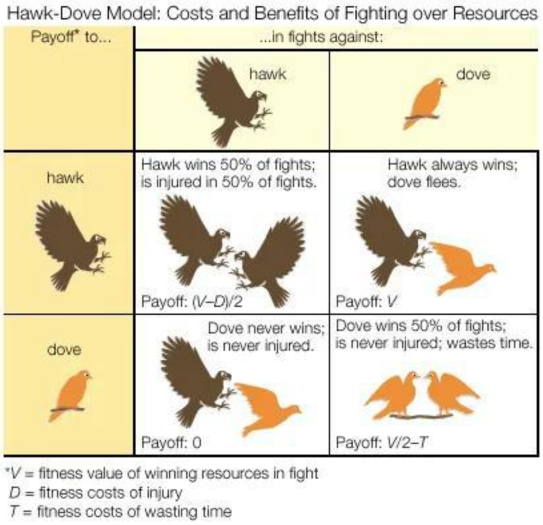


A higher number of EEG low-frequency THETA waves was associated with a relative impairment in impulse control.

But returning to the previous study which we talked about (the girl one), in that it was also done an EEG study, and what it was shown that the exact same thing; when the subject failed to suppress the emotional response, they had slow wave episodes within the pre-frontal cortex, Failure to control emotional responses was associated to onset of EEG sleep-like activity in the parieto-temporal frontomedial insula network.

Another study wanted instead to see how much we can control our instinct when we are tired. In what extend brain fatigue may affect social behaviour?

In the Ordini study they had 2 groups of study, doing economic games*, and one of the two went for a series of cognitively tiring tasks. Both then went to do the game, and they all had EEG recording on their head. You see what you would expect, the ones which were tired had a lot of slow wave activity in the pre-frontal cortex, but what is very interesting is how they performed in the game, one which was designed in a way that if you persist choosing the aggressive solution on the long run you lose money and resources. People who performed at this game without being fatigued, choose 86% of the time the pacific solution, on the contrary people cognitively engaged in a massive way for an hour, would choose the pacific solution only 41% of the time. So, if you make the choice while you are a little tired (1 hour is not so much), this does affect the choice we make, we go towards an aggressive choice even if it is counterproductive.



*Games which have been used to understand how people behave in social interactions, for example the ultimatum game (there are two players, and you have the opportunity to split or accept an amount of money, etc..). If they offer to split in 9 and 1, from a rational point of view saying no to this is not justified, because you both lose money. What makes you answer in this way is the unfairness you perceive in the personal decision.

Implications for everyday social life and work

- Brain Fatigue appears after a relative short time of cognitive engagement
- Brain Fatigue affects task-related areas and impairs performance
- Prolonged self-control leads to Frontal Fatigue
- Frontal fatigue favors aggressive behavior in social interactions
- Overall, these findings carry implications for scheduling work meetings, school teaching or judicial mediations
- Science appears to provide support to popular wisdom says like "
    - *sleep on it – dormici sopra*
    - *the night brings counsel – la notte porta consiglio*
    - *The early bird catches the worm - la mattina ha l'oro in bocca*

This has implications for everyday life, and also for forensic psychiatric and the law. How do we control impulse and aggressive behaviour?

Mental capacity for the law. In order for someone to behave in a responsible way you need to have:

- Capacity to understand: The ability to understand the nature of one own's actions and their consequences.

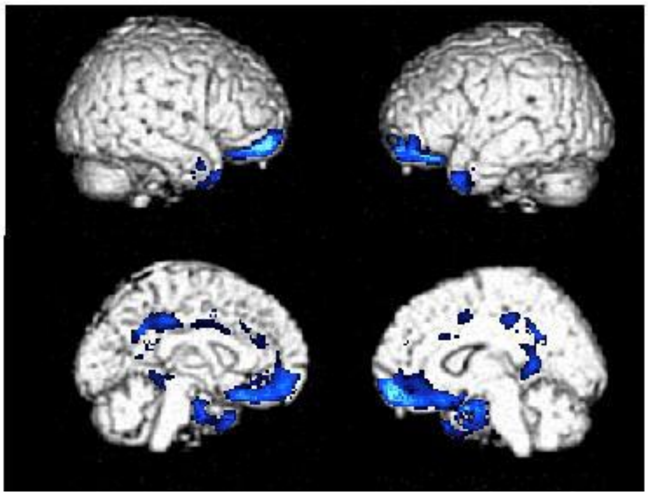
- Capacity to will: The ability to control one own's behaviour.

If either one of these capacities are lost, you are not legally responsible.

Asking individuals to imagine becoming aggressive, you have the same brain response as if you are really being it. In the ventromedial pre-frontal cortex, you have a functional suppression (reduction in blood flow) which accompanies the elicitation at the imagining level of the aggressive behaviour. Even imagining becoming aggressive there is a shutdown of the pre-frontal cortex, which should normally control the release of aggressive behaviour.

Years later, a study demonstrated that if you compare psychopathic criminals with a controlled group in terms of brain structure (neuronal density), you can see two things in the differential graph:

- Differences are not really across the whole brain, is essentially identical with the exception of the ventromedial pre-frontal cortex, in which psychopathic criminals have a fewer number of neurons.
- There is a shut down in the ventromedial cortex.



Light blue indicates the functional shut down, the darker blue indicates a structural difference, with a reduction in the number of neurons.